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## SCHOOL OF COMPUTER SCIENCE AND ENGINEERING (SCOPE)

# LAB WEEK - 2

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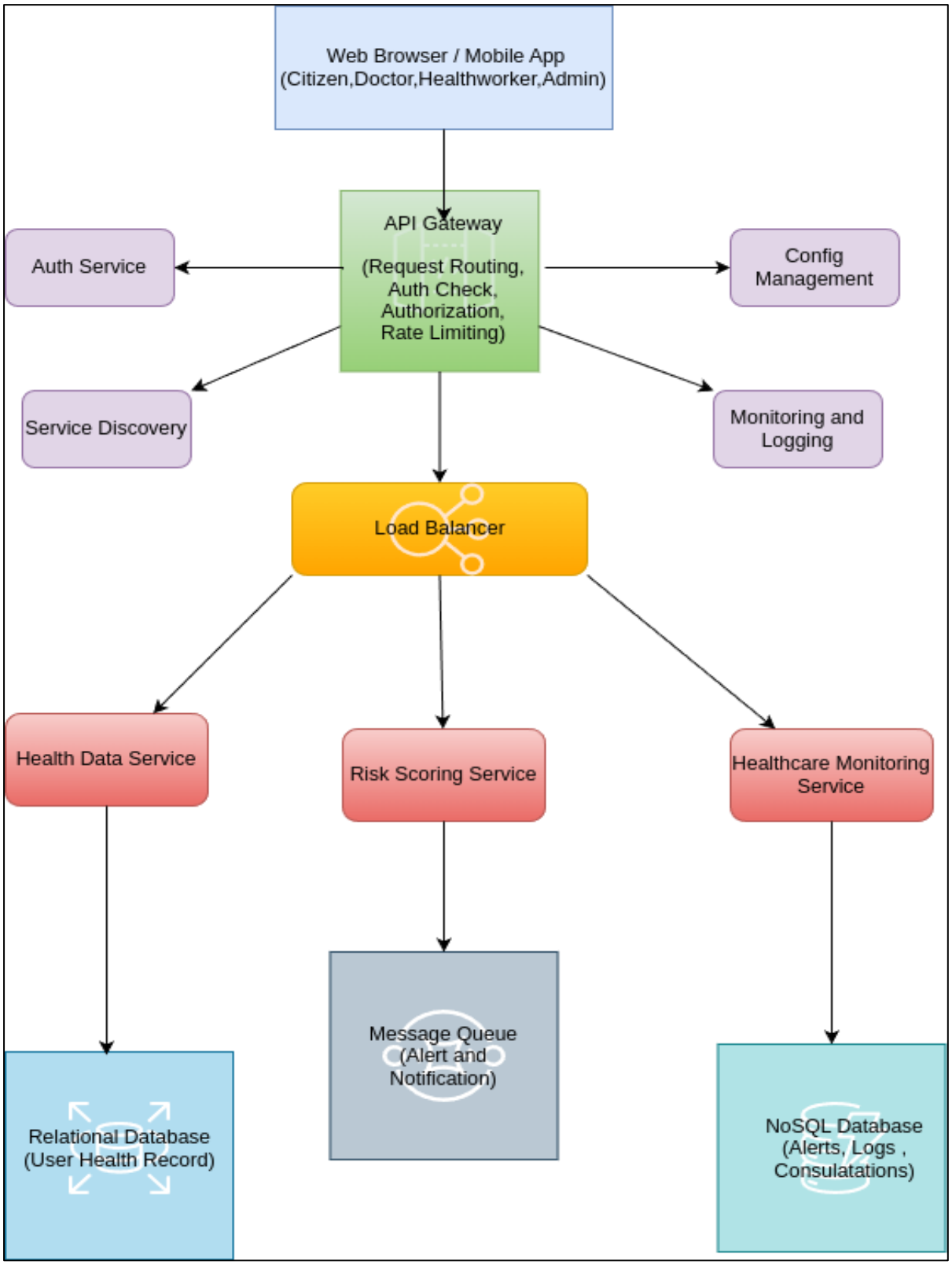
## ARCHITECTURE DIAGRAM



# SRS

## (Software Requirements Specification)

## 1. Introduction

### 1.1 Purpose

This Software Requirements Specification (SRS) document provides a complete and detailed description of the **Community Micro-Health Monitoring & Alert System**. It specifies the system's functional and non-functional requirements, intended users, constraints, and interfaces. This document serves as a reference for developers, testers, project evaluators, and faculty members to clearly understand what the system is expected to do.

### 1.2 Scope

The Community Micro-Health Monitoring & Alert System is a web-based application designed to enable early detection of health issues at the community level. Citizens can report basic daily health parameters, the system evaluates risk levels using predefined rules, and alerts are automatically generated for health workers and doctors. The system aims to improve preventive healthcare, reduce response time, and support data-driven health monitoring.

### 1.3 Definitions, Acronyms, and Abbreviations

- **Citizen**: A registered community member submitting health data
- **Health Worker**: A local healthcare official monitoring alerts
- **Doctor**: A medical professional reviewing cases and providing guidance
- **Admin**: System administrator
- **SRS**: Software Requirements Specification
- **API**: Application Programming Interface

### 1.4 References

- IEEE 830 / IEEE 29148 SRS Standards
- Community Health Monitoring Guidelines

### 1.5 Document Overview

This document contains an overall system description, functional and non-functional requirements, system interfaces, constraints, and future enhancements.

## 2. Overall Description

### 2.1 Product Perspective

The system is a standalone, web-based application built using a service-oriented or microservices architecture. It interacts with users through web browsers and internally uses backend services for data processing, alert generation, and monitoring.

### 2.2 Product Functions

- User registration and authentication
- Daily health data collection
- Risk level calculation (low, medium, high)
- Alert generation and notification
- Health report viewing and consultation
- Administrative monitoring and management

### 2.3 User Classes and Characteristics

- **Citizens**: Basic digital literacy, submit daily health data
- **Health Workers**: Monitor alerts and guide citizens
- **Doctors**: Review reports and provide medical advice
- **Administrators**: Manage users and system data

### 2.4 Operating Environment

- Web browsers such as Chrome, Firefox, or Edge
- Server hosted on cloud or local infrastructure
- Database servers for structured and unstructured data

### 2.5 Design and Implementation Constraints

- Requires continuous internet connectivity
- Must follow basic data privacy and security practices
- Limited to non-invasive health parameters

### 2.6 Assumptions and Dependencies

- Users provide accurate health information

- Health workers respond to alerts in a timely manner
- System availability depends on server uptime

## 3. Functional Requirements

### 3.1 User Registration and Authentication

- The system shall allow users to register using basic personal information
- The system shall authenticate users based on credentials
- The system shall support role-based access (Citizen, Doctor, Health Worker, Admin)

### 3.2 Health Data Submission

- The system shall allow citizens to submit daily health parameters
- The system shall validate input values before storing
- The system shall store submitted data securely

### 3.3 Risk Assessment

- The system shall evaluate submitted data using predefined rules
- The system shall assign a risk level: low, medium, or high
- The system shall store risk evaluation results

### 3.4 Alert Generation and Notification

- The system shall generate alerts for medium and high-risk cases
- The system shall notify assigned health workers
- The system shall support asynchronous alert delivery

### 3.5 Doctor and Health Worker Access

- Doctors shall be able to view detailed health reports
- Health workers shall monitor alerts for their locality
- Doctors shall provide consultation or recommendations

### 3.6 Administrative Functions

- Admins shall manage user accounts

- Admins shall view community health trends
- Admins shall monitor system usage and logs

## 4. Non-Functional Requirements

### 4.1 Performance Requirements

- The system shall process health submissions within acceptable time limits
- Alerts shall be generated with minimal delay

### 4.2 Security Requirements

- User data shall be protected against unauthorized access
- Secure authentication and authorization shall be enforced

### 4.3 Usability Requirements

- The system shall have an intuitive and user-friendly interface
- The system shall be accessible on desktop and mobile browsers

### 4.4 Reliability Requirements

- The system shall operate continuously with minimal downtime
- Data loss shall be prevented using backups

### 4.5 Scalability Requirements

- The system shall support an increasing number of users
- Architecture shall allow easy addition of new features

## 5. System Interfaces

### 5.1 User Interface

- Web-based forms for health data entry
- Dashboards for doctors, health workers, and admins

### 5.2 Hardware Interface

- No specific hardware requirements

### 5.3 Software Interface

- Database management systems
- Notification services (email/SMS)

### 5.4 Communication Interface

- REST-based APIs for service communication

## 6. Other Requirements

- System activity logs shall be maintained
- Periodic data backups shall be performed

## 7. Future Enhancements

- Integration with wearable health devices
- AI-based predictive analytics
- Mobile application support

## 8. Conclusion

The Community Micro-Health Monitoring & Alert System provides a structured and reliable approach to community health monitoring. By enabling early detection, automated alerts, and effective coordination between citizens and healthcare professionals, the system contributes significantly to preventive healthcare and improved public health outcomes.